Akoizo Original Practice - Form 3 - Answer key

Division B | Disease Detectives | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) A line list usually has one row per…

Unit: Outbreak investigation and descriptive evidence.
Worked-case data: Six cases have onset on day 1, eight on day 2, and two on day 3. Their report dates are all day 5.

A. Case or explicitly defined event

B. Map color

C. Hypothesis only

D. Calendar month necessarily

Correct: A

2. (2 points) Directly contacting facilities for case records is…

Unit: Surveillance and the public health information cycle.
Worked-case data: Reported cases double in the month a new laboratory and more testing sites begin reporting.

A. Active surveillance

B. Only passive reporting

C. A randomized trial

D. A case-control definition

Correct: A

3. (2 points) Rehabilitation after established disability is usually…

Unit: Levels of prevention and population strategies.
Worked-case data: In 1000 synthetic people, 10 have a condition. A test has sensitivity 90% and specificity 90%. Calculate expected counts and PPV.

A. Tertiary prevention

B. Primary prevention

C. A new incidence definition

D. A case-control selection rule

Correct: A

4. (2 points) What can create an artificial report-date spike?

Unit: Outbreak investigation and descriptive evidence.
Worked-case data: Six cases have onset on day 1, eight on day 2, and two on day 3. Their report dates are all day 5.

A. Batch reporting

B. Perfect onset data

C. A uniform denominator

D. Removing duplicate cases

Correct: A

5. (2 points) A selected network of sites is…

Unit: Surveillance and the public health information cycle.
Worked-case data: Reported cases double in the month a new laboratory and more testing sites begin reporting.

A. Sentinel surveillance

B. A complete census necessarily

C. A perfect random sample automatically

D. No surveillance

Correct: A

6. (2 points) Why can PPV be low despite high specificity?

Unit: Levels of prevention and population strategies.
Worked-case data: In 1000 synthetic people, 10 have a condition. A test has sensitivity 90% and specificity 90%. Calculate expected counts and PPV.

A. Low prevalence can leave many false positives

B. Sensitivity equals prevalence

C. All tests are random

D. PPV ignores false positives

Correct: A

7. (2 points) A spot map directly supplies…

Unit: Outbreak investigation and descriptive evidence.
Worked-case data: Six cases have onset on day 1, eight on day 2, and two on day 3. Their report dates are all day 5.

A. Case locations

B. Population-adjusted risk automatically

C. The causal agent

D. Complete exposure histories

Correct: A

8. (2 points) A broad syndrome signal often trades specificity for…

Unit: Surveillance and the public health information cycle.
Worked-case data: Reported cases double in the month a new laboratory and more testing sites begin reporting.

A. Timeliness or sensitivity

B. Guaranteed causality

C. Perfect diagnosis

D. Elimination of false alarms

Correct: A

9. (2 points) A safe-water intervention before illness is…

Unit: Levels of prevention and population strategies.
Worked-case data: In 1000 synthetic people, 10 have a condition. A test has sensitivity 90% and specificity 90%. Calculate expected counts and PPV.

A. Primary prevention

B. Tertiary prevention necessarily

C. Only surveillance

D. A diagnostic test

Correct: A

10. (4 points) Give three useful line-list fields.

Unit: Outbreak investigation and descriptive evidence.
Worked-case data: Six cases have onset on day 1, eight on day 2, and two on day 3. Their report dates are all day 5.

answer: Examples: onset date, standardized symptoms, exposure history, location, and confirmation status. (4 points).

11. (4 points) List four steps in a surveillance cycle.

Unit: Surveillance and the public health information cycle.
Worked-case data: Reported cases double in the month a new laboratory and more testing sites begin reporting.

answer: Ongoing collection, analysis, interpretation, and dissemination for action, with feedback. (4 points).

12. (4 points) Calculate expected false positives in the example.

Unit: Levels of prevention and population strategies.
Worked-case data: In 1000 synthetic people, 10 have a condition. A test has sensitivity 90% and specificity 90%. Calculate expected counts and PPV.

answer: 990 x 0.1 = 99. (4 points).

13. (4 points) Why vary epidemic-curve bin width?

Unit: Outbreak investigation and descriptive evidence.
Worked-case data: Six cases have onset on day 1, eight on day 2, and two on day 3. Their report dates are all day 5.

answer: To assess whether apparent waves or peaks depend on arbitrary grouping rather than a robust temporal pattern. (4 points).

14. (4 points) Why inspect testing changes?

Unit: Surveillance and the public health information cycle.
Worked-case data: Reported cases double in the month a new laboratory and more testing sites begin reporting.

answer: More or different testing can change reported counts independently of true disease occurrence. (4 points).

15. (4 points) Calculate PPV.

Unit: Levels of prevention and population strategies.
Worked-case data: In 1000 synthetic people, 10 have a condition. A test has sensitivity 90% and specificity 90%. Calculate expected counts and PPV.

answer: 9/(9+99) = 9/108 about 8.3%. (4 points).

16. (4 points) Why exclude suspected exposure from the outcome definition?

Unit: Outbreak investigation and descriptive evidence.
Worked-case data: Six cases have onset on day 1, eight on day 2, and two on day 3. Their report dates are all day 5.

answer: Including it would make exposure and case status dependent by construction and bias the association test. (4 points).

17. (4 points) What does representativeness concern?

Unit: Surveillance and the public health information cycle.
Worked-case data: Reported cases double in the month a new laboratory and more testing sites begin reporting.

answer: Whether observed cases and covered populations adequately reflect the relevant person-place-time distribution. (4 points).

18. (4 points) Why is finding more cases not automatically a benefit?

Unit: Levels of prevention and population strategies.
Worked-case data: In 1000 synthetic people, 10 have a condition. A test has sensitivity 90% and specificity 90%. Calculate expected counts and PPV.

answer: Detection must lead to useful outcomes and be weighed against false results, overdiagnosis, access, and intervention effects. (4 points).